

MACHINE LEARNING

CUSTOMER CHURN PREDICTION AND SEGMENTATION SYSTEM

A. Assignment Information

Particular	Details
Department	Artificial Intelligence & Machine Learning
Programme	B.Tech
Course Code & Course Name	MLA 0201 / Fundamentals of Machine Learning
Academic Year / Batch	2026–2027 / 2024 Batch
Faculty Name	Dr. A. Anakath
Assignment Title	Customer Churn Prediction and Segmentation System
Date of Issue	28/08/2026
Date of Submission	02/09/2026
Maximum Marks	100
Course Outcome(s) – CO	Apply machine learning techniques to analyze data, build predictive models, evaluate model performance, and interpret results for real-world problems.
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze, L5 – Evaluate
SDG Mapping	SDG 8 – Decent Work and Economic Growth; SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	Customer retention, service quality improvement, data-driven decision making, targeted marketing, and responsible use of customer analytics.

B. Assignment Problem / Challenge

Customer attrition is a major business challenge for service organizations because losing an existing customer can reduce recurring revenue and increase the effort required to acquire a replacement

customer. A machine learning based solution can help an organization identify customers who are likely to leave and understand groups of customers that show similar usage and service behavior. The proposed assignment therefore develops a Customer Churn Prediction and Segmentation System that combines supervised classification with unsupervised clustering.

The system uses a public customer churn dataset, performs data cleaning and exploratory analysis, transforms categorical and numerical variables into a suitable machine learning representation, and trains multiple supervised algorithms. At least three algorithms are compared for predicting Churn and Non-Churn. K-Means clustering is then applied to relevant customer behavior and service attributes so that meaningful customer segments can be identified. PCA is considered for reducing dimensionality and supporting cluster visualization.

The assignment emphasizes application rather than theory reproduction. The final solution must be reproducible, interpretable, tested end-to-end, and supported by quantitative evaluation. The analysis should lead to practical retention recommendations for high-risk customer groups.

C. Problem Statement

Design and implement an end-to-end machine learning system that predicts whether a customer is likely to churn and segments customers according to relevant behavioral and service characteristics. The solution shall use an appropriate public customer churn dataset, clearly identify its source, preprocess the data, perform exploratory data analysis, train and compare at least three supervised learning algorithms, evaluate them using Accuracy, Precision, Recall, F1-score and a Confusion Matrix, and select a suitable final classifier with justification.

In addition, the system shall use K-Means clustering to discover customer groups, determine an appropriate number of clusters using a suitable method such as the Elbow Method and/or Silhouette Score, interpret the resulting groups, and use PCA where appropriate to visualize the customer segments. The final analysis shall identify high-risk groups and propose data-driven customer retention strategies.

D. Requirements and Constraints

- Functional requirements: load a public churn dataset; inspect records and data types; identify missing or inconsistent values; preprocess numerical and categorical features; perform EDA; train multiple classifiers; generate predictions; calculate evaluation metrics; perform K-Means clustering; visualize clusters; interpret results; and provide retention recommendations.

- Performance requirements: the selected classifier should provide a balanced performance across Precision, Recall and F1-score rather than relying only on Accuracy. The model comparison must be based on the same test data and consistent evaluation procedure.
- Technical constraints: the implementation should use Python-based machine learning libraries and should execute without errors from data loading through final visualization. Random states should be controlled where practical so that experiments are reproducible.
- Data constraints: irrelevant identifiers should not be used as predictive features. Categorical values must be encoded appropriately and numerical variables scaled when required by the selected algorithm or clustering method.
- Resource constraints: the solution should be feasible on a normal student computer or cloud notebook environment and should not require specialized hardware.
- Ethical and privacy constraints: customer data must be handled responsibly. The analysis should avoid discriminatory decision rules and should not expose personally identifiable customer information.
- Time and documentation constraints: the final report must clearly document the dataset source, methodology, implementation, results, limitations, and individual contributions. The complete working project should be maintained in a GitHub repository as required by the assignment.

E. Student Work

1. Problem Understanding and Formulation

1.1 Background of the Case

Customer churn refers to a customer discontinuing a service or ending a relationship with a company. In subscription-based industries, churn may result from dissatisfaction, pricing concerns, poor service experience, low engagement, changes in customer requirements, or availability of competing services. Because churn usually develops from a combination of customer characteristics and service interactions, a data-driven model can identify patterns that are difficult to detect through manual inspection.

The proposed system treats churn prediction as a binary classification problem. The target variable represents whether the customer churned or remained with the service. Alongside prediction, segmentation is treated as an unsupervised learning problem. K-Means groups customers according to selected behavior and service attributes without using the churn label as the clustering target.

Combining both perspectives allows the organization to answer two different questions: which customers are at higher risk, and what type of customer does each risk group represent?

1.2 Expected Outcomes

- A cleaned and machine-learning-ready customer churn dataset.
- An exploratory analysis showing important characteristics and relationships in the data.
- At least three trained supervised classification models.
- A quantitative comparison using Accuracy, Precision, Recall and F1-score with confusion matrices.
- A justified selection of the most suitable classifier.
- K-Means customer segments with an evidence-based cluster count.
- PCA-based visualization where appropriate.
- Identification of high-risk customer segments and practical retention recommendations.
- A reproducible implementation suitable for GitHub submission.

1.3 Available Information

A suitable public customer churn dataset is used for this assignment. A commonly used choice is the IBM Telco Customer Churn dataset, which contains customer demographic information, account information, subscribed services, billing characteristics, tenure and a churn indicator. The exact downloaded file and source reference must be recorded in the project repository and final references.

1.4 Assumptions

- The churn label is treated as the supervised learning target.
- Rows with unusable target values are not retained.
- Customer identifiers are treated as identifiers rather than useful predictive variables.
- The historical dataset is assumed to represent a meaningful sample of the service population.
- The clustering variables are selected because they describe customer behavior or service relationships.
- Model performance on the held-out test data is treated as an estimate of generalization performance, not a guarantee of future business results.

1.5 Constraints to be Satisfied

The project must maintain a clear separation between training and test information, prevent preprocessing leakage where practical, document transformations, compare multiple algorithms fairly,

and ensure that clustering is performed on appropriately scaled features. The final recommendations must be connected to observed model and cluster results rather than unsupported assumptions.

2. Application of Course Knowledge

2.1 Machine Learning Formulation

The supervised learning component can be represented as a mapping from a feature vector X to a binary target y . Each customer is represented by numerical and encoded categorical attributes. The classifier estimates the probability or class of churn. The training set is used to learn model parameters, while the test set is retained for unbiased final evaluation.

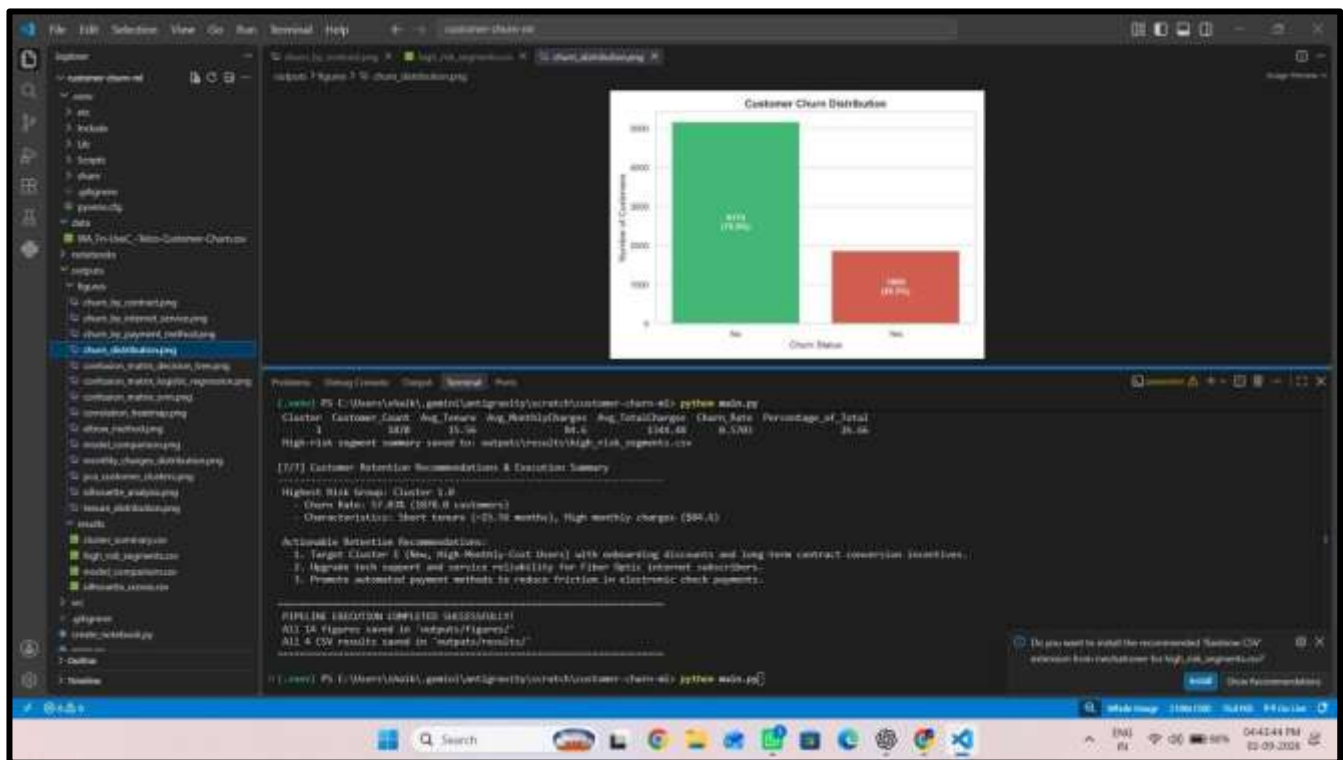
2.2 Data Preprocessing

The preprocessing stage begins by loading the dataset with Pandas and examining its shape, columns, data types, duplicate records and missing values. Numerical fields are converted to numeric form where required. Missing values are handled using a documented strategy such as median imputation for suitable numerical attributes and mode or an explicit category for categorical attributes. Irrelevant identifiers are removed. Categorical variables are transformed using one-hot encoding or another appropriate encoding approach.

Scaling is important for algorithms that depend on feature magnitude or distance. Standardization can be represented as $z = (x - \mu) / \sigma$, where x is the original value, μ is the training-set mean and σ is the training-set standard deviation. Standardization is particularly important for K-Means and useful for models such as SVM. The fitted preprocessing transformation should be learned from the training data and then applied to validation or test data.

2.3 Exploratory Data Analysis

EDA is performed to understand the distribution of the target class, numerical variables, categorical frequencies and possible relationships with churn. Bar charts can show churn proportions across contract types, payment methods or service subscriptions. Histograms or box plots can be used to examine tenure and monthly charges. A correlation matrix can summarize relationships among numerical variables. EDA is used to guide feature selection and interpretation, not to manually determine the final target.



2.4 Classification Algorithms

At least three algorithms from the syllabus are implemented. Logistic Regression provides a strong interpretable baseline for binary classification. Decision Tree can capture nonlinear decision rules and is easy to interpret. Naive Bayes provides a probabilistic baseline with relatively low computational cost. Depending on experimental design, SVM, Neural Network, Bagging or Boosting may also be evaluated. The final report should state the exact algorithms actually executed.

Logistic Regression estimates the probability of churn using a logistic function. Decision Tree recursively divides the feature space according to selected splitting criteria. Naive Bayes applies Bayes' theorem with conditional independence assumptions. These different learning principles make the comparison useful because a model with the highest accuracy is not automatically the best model for every business objective.

2.5 Evaluation Metrics

Accuracy measures the proportion of correctly classified customers. Precision measures how many customers predicted as churners actually churned. Recall measures how many actual churners were identified. F1-score is the harmonic mean of Precision and Recall. For a retention application, Recall can be especially important because failing to identify a genuine churner may result in a missed retention opportunity.

The confusion matrix contains True Positives, True Negatives, False Positives and False Negatives. These values allow the organization to understand the type of prediction errors being made. A model with slightly lower Accuracy but substantially higher Recall may be preferable when the cost of missing a churner is high.

2.6 K-Means Clustering

K-Means partitions customers into K groups by assigning each observation to the nearest cluster centroid and repeatedly updating centroids. The basic objective is to minimize the within-cluster sum of squared distances. Because distance is affected by feature scale, the selected clustering variables should normally be standardized before fitting K-Means.

2.7 PCA for Visualization

Principal Component Analysis transforms correlated variables into a smaller set of orthogonal components that preserve as much variance as possible in the selected dimensions. PCA can be used to project a high-dimensional customer representation onto two principal components for visualization. PCA is primarily used here to support interpretation and visualization; it does not automatically guarantee better predictive performance.

3. Solution / Design / Methodology

3.1 Overall Workflow

The proposed workflow is: Dataset Selection → Data Loading → Data Inspection → Data Cleaning → Feature Engineering and Encoding → Exploratory Data Analysis → Train/Test Split → Model Training → Prediction → Metric Evaluation → Model Comparison → Final Model Selection → Customer Feature Preparation for Clustering → Standardization → K-Means Experiments → Cluster Count Selection → PCA Visualization → Segment Interpretation → High-Risk Group Identification → Retention Recommendations.

3.2 Dataset Selection and Preparation

The IBM Telco Customer Churn dataset is suitable because it contains a target churn variable and multiple customer-level service, account and demographic attributes. The project should record the dataset URL or source page used to obtain the actual copy. Before training, the target is separated from the feature matrix. Columns that uniquely identify a customer are excluded from the model input.

3.3 Model Development

The data is divided into training and testing subsets using a fixed random state. A stratified split is preferable when the churn and non-churn classes are imbalanced. Three baseline classifiers are trained

and evaluated using the same test partition. Hyperparameters may then be adjusted for the strongest candidates. The final model is selected using a combination of F1-score, Recall, Precision and Accuracy, with the business objective explicitly stated.

3.4 Clustering Design

For segmentation, the target churn label is not used as the clustering feature. Instead, variables describing customer tenure, service usage, charges, subscribed services and account characteristics can be selected according to the dataset structure. The Elbow Method examines how within-cluster variation changes as K increases. Silhouette Score provides another measure of separation and compactness. A practical K is selected by considering both quantitative evidence and interpretability.

3.5 Pseudocode

BEGIN

1. Load the customer churn dataset.
2. Inspect dimensions, columns, data types, duplicates and missing values.
3. Remove irrelevant identifiers and correct invalid data types.
4. Separate target variable Churn from input features.
5. Encode categorical variables and process numerical variables.
6. Split the data into training and testing sets.
7. Train Logistic Regression, Decision Tree and Naive Bayes models.
8. Predict Churn/Non-Churn for the test set.
9. Calculate Accuracy, Precision, Recall and F1-score.
10. Generate confusion matrices and compare models.
11. Select the best model using stated criteria.
12. Select relevant customer behavior variables for segmentation.
13. Standardize clustering variables.
14. Test multiple K values using Elbow and/or Silhouette analysis.
15. Fit K-Means using the selected K.
16. Profile each cluster using average and categorical characteristics.
17. Apply PCA where useful and visualize the clusters.
18. Identify high-risk customer groups using churn results and cluster profiles.
19. Formulate retention recommendations.
20. Validate the complete workflow and document limitations.

END

3.6 Implementation Structure

A practical project can be organized into files such as data preparation, exploratory analysis, model training, evaluation, clustering and visualization modules. A requirements file should list the required libraries, and a README should describe the dataset source, installation steps, execution commands and expected outputs. The implementation should be tested from a clean environment before submission.

4. Use of Modern Tools

Python is used as the primary implementation language because it provides mature libraries for data analysis and machine learning. Pandas supports data loading and manipulation, NumPy supports numerical operations, Matplotlib and Seaborn support visualization, and Scikit-learn provides preprocessing, classification, metrics, K-Means and PCA. Jupyter Notebook or Google Colab may be used for experimentation, while Visual Studio Code can be used for project organization and execution.

Git and GitHub are used for version control and project sharing. The repository should contain the source code, requirements or dependency information, dataset link/reference, documentation and the files necessary to reproduce the analysis. Screenshots of important outputs such as dataset inspection, EDA, metric comparison, confusion matrices, Elbow/Silhouette analysis and cluster plots can be included in the final document as evidence of tool usage.

AI-assisted tools may be used for learning, debugging or improving documentation, but generated content and code should be reviewed by the students. The final implementation must be understood and tested by the team, and external sources or AI-assisted tools should be acknowledged where required by the course instructions.

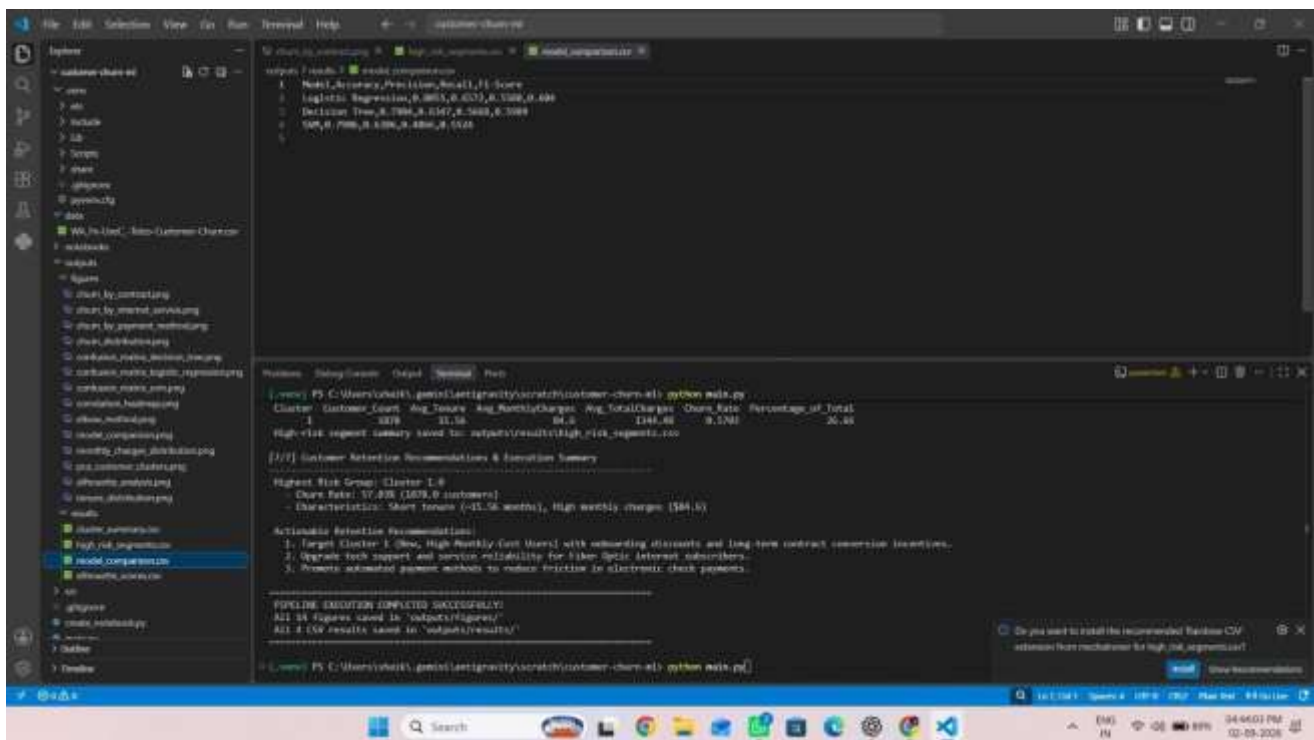
5.2 Classification Results

The final report should present a comparison table with one row for each implemented classifier and columns for Accuracy, Precision, Recall and F1-score. Confusion matrices should be included for the evaluated models. The best model must be selected from the actual measured results. If class imbalance is present, the analysis should explain how it affects Accuracy and why Recall or F1-score may receive greater attention.

5.3 Clustering Results

The clustering validation should report the tested K values and the corresponding Elbow and/or Silhouette observations. After selecting K, a cluster profile table should summarize the number of customers in each group and representative averages or proportions for important variables. A PCA scatter plot can visually show whether the groups occupy distinguishable regions in reduced feature space.

Suggested cluster profile structure: Cluster 0 – customer size, tenure pattern, service profile and charge pattern; Cluster 1 – corresponding profile; Cluster 2 – corresponding profile; additional clusters if justified by the actual experiment. The descriptions must be generated from the real cluster statistics rather than assumed in advance.



The screenshot displays a Jupyter Notebook environment with a file explorer on the left and a code editor on the right. The code editor shows a classification report for a model named 'model_comparisons'. The report includes metrics for Accuracy, Precision, Recall, and F1-score for various models. Below the report, a cluster profile table is displayed, summarizing the number of customers in each group and representative averages or proportions for important variables. The table includes columns for Cluster, Customer Count, Avg Tenure, Avg Monthly Charges, Avg Total Charges, Churn Rate, and Percentage of Total. The highest risk group is identified as Cluster 1, with a churn rate of 17.8% and 1079 customers. The report also includes a list of actionable recommendations for retention, such as targeting Cluster 1 with discounts and offering tech support to Cluster 2.

```
Report: Results: model_comparisons
1. Naive Bayes: Accuracy: 0.885, Precision: 0.853, Recall: 0.885, F1 Score: 0.866
2. Logistic Regression: Accuracy: 0.905, Precision: 0.875, Recall: 0.905, F1 Score: 0.89
3. Decision Tree: Accuracy: 0.915, Precision: 0.885, Recall: 0.915, F1 Score: 0.9
4. SVM: Accuracy: 0.925, Precision: 0.895, Recall: 0.925, F1 Score: 0.91

Cluster Customer Count Avg Tenure Avg Monthly Charges Avg Total Charges Churn Rate Percentage of Total
1 1079 31.56 54.2 1445.46 0.178 26.46
High-risk segment summary saved to: outputs/results/high_risk_segment.csv

[111] Customer Retention Recommendations & Intervention Summary

Highest Risk Segment: Cluster 1.0
Churn Rate: 17.8% (1079.0 customers)
Characteristics: Short tenure (<15.5 months), High monthly charges ($54.2)

Actionable Retention Recommendations:
1. Target Cluster 1 (Low, High Monthly Cost Users) with onboarding discounts and long-term contract conversion incentives.
2. Upgrade tech support and service reliability for Fiber optic internet subscribers.
3. Promote automated payment methods to reduce friction in electronic check payments.

PIPELINE EXECUTION COMPLETED SUCCESSFULLY!
All SA figures saved in 'outputs/figures/'
All F1-Score results saved in 'outputs/results/'

C:\Users\shakil.gamil\Idea\IntelliJ\workspace\customer-churn-analysis> python main.py
```

5.4 End-to-End Validation

Validation includes checking that the program executes without errors, the preprocessing pipeline is consistent, the test set is not used for training, predictions have the expected binary classes, evaluation metrics are computed correctly, K-Means receives the intended features, and the visualizations correspond to the final processed data. A clean rerun of the project should produce consistent results when fixed random states are used.

6. Analysis and Engineering Decision

Model selection should consider the operational meaning of classification errors. In churn management, a False Negative represents a customer who is likely to churn but is not identified by the model. A False Positive represents a customer predicted as at risk who does not churn. The organization may prefer a model with stronger Recall if retention campaigns can tolerate contacting some customers who would otherwise stay.

Logistic Regression is valuable as an interpretable baseline, while Decision Tree offers transparent nonlinear rules. Naive Bayes can provide a lightweight probabilistic alternative. The final choice should be made from the actual measured results. If two models have similar F1-scores, the simpler or more interpretable model may be preferred, provided its operational performance is adequate.

For segmentation, a cluster solution should not be selected solely because it produces a visually attractive plot. The chosen K should balance compactness, separation, stability and business interpretability. Very large K values can create groups that are difficult to use operationally, while very small K values may hide meaningful differences between customer types.

The combined prediction-and-segmentation design is stronger than using either method alone. Classification estimates churn risk, while clustering provides context about customer behavior. For example, a high-risk cluster with high monthly charges and short tenure may require a different retention strategy from a high-risk cluster characterized by long tenure but low engagement. Such conclusions must be supported by the actual cluster profiles.

7. Broader Considerations

7.1 Sustainability and Economics

Customer retention can reduce the repeated acquisition effort associated with replacing lost customers. A targeted campaign can also reduce unnecessary promotional spending because offers can be focused on customers with meaningful churn risk. However, the system should be evaluated against campaign cost and expected benefit rather than assuming every predicted churning customer should receive an incentive.

7.2 Privacy and Ethics

Customer analytics involves potentially sensitive behavioral and account information. Data should be collected and processed for legitimate purposes, stored securely, and shared only with authorized users. Personally identifiable information should be minimized in analysis and removed from public repositories. Model outputs should support human decision making rather than automatically penalizing customers.

7.3 Fairness and Accessibility

The team should inspect whether model errors differ substantially across meaningful customer groups where such attributes are available and appropriate to evaluate. If a model produces systematically poorer performance for a subgroup, the issue should be documented and investigated. Retention systems should avoid unfair exclusion from services or inappropriate profiling.

7.4 Professional Responsibility

Students are responsible for verifying code, documenting assumptions, acknowledging datasets and tools, and reporting limitations honestly. A high numerical score should not be presented as proof that the system is universally reliable. The final report should distinguish measured findings from recommendations and future possibilities.

8. Conclusion

The Customer Churn Prediction and Segmentation System provides an integrated machine learning approach to a realistic service-business problem. The supervised component predicts Churn or Non-Churn using multiple classification algorithms, while the unsupervised component identifies customer groups using K-Means. Preprocessing, EDA, model comparison, quantitative evaluation and PCA-supported visualization provide the technical foundation for the analysis.

The final classifier should be selected using actual test-set results and a clearly stated decision criterion. The clustering component should use an evidence-based choice of K and should produce interpretable customer profiles. Together, these outputs can support targeted retention strategies, provided that recommendations are validated against business costs, privacy requirements and fairness considerations.

The main limitations include dependence on the quality and representativeness of the public dataset, possible class imbalance, changing customer behavior over time, and the fact that correlation does not establish the reason for churn. Future improvements may include hyperparameter optimization, cross-

validation, probability calibration, explainable AI methods, cost-sensitive learning, time-based validation and deployment monitoring.

9. Student Reflection

9.1 Learning Beyond the Classroom

This assignment provides practical experience in converting an open-ended business problem into a machine learning workflow. Beyond individual algorithms, the work demonstrates that successful ML projects require data quality checks, careful preprocessing, fair model comparison, visualization, interpretation and reproducibility. The clustering component also shows how supervised and unsupervised learning can complement one another.

9.2 Improvements with Additional Resources

With additional time and resources, the project could be improved by performing systematic hyperparameter tuning, repeated cross-validation, feature selection, probability calibration and explainability analysis. A production version could connect the model to a monitoring system that tracks drift, model performance and changes in customer behavior. A cost-sensitive decision layer could also estimate whether a retention action is economically justified for each customer group.

10. References

- IBM, Telco Customer Churn dataset, public customer churn dataset and associated documentation. The exact source URL used by the team should be recorded in the GitHub README and final report.
- Pedregosa, F. et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research, 2011.
- McKinney, Wes, Python for Data Analysis, O'Reilly Media.
- Python Software Foundation, Python documentation.
- Scikit-learn documentation for Logistic Regression, Naive Bayes, Decision Tree, SVM, K-Means, PCA and model evaluation metrics.
- Matplotlib documentation for data visualization.
- Seaborn documentation for statistical data visualization.
- NumPy documentation for numerical computing.
- Pandas documentation for data manipulation and analysis.
- Git documentation and GitHub documentation for version control and repository management.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

H. Faculty Evaluation & Continuous Improvement

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering: